Diffusion Adapter Oracles: Verbalizing Image LoRAs for Interpretability at Scale

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Abstract

We introduce the diffusion adapter oracle (DAO), a meta-model that reads the weights of an image diffusion transformer's LoRA adapter and says in natural language what that adapter does, without running it. A frozen encoder converts each module's low-rank update into a symmetry-preserving token and injects the token sequence into a language-model interpreter that operates in a different architecture and residual space from the diffusion model; only interpreter-side parameters are trained. DAO names the concept of 58 of 105 held-out adapters (55.2%, 85.6 times chance), against 14 of 105 (13.3%) for nearest-neighbor retrieval over the same tokens and 0 of 105 for a matched interpreter trained on shuffled tokens. As a causal check, substituting another adapter's tokens at inference drops exact match to 0 of 24 at the two largest budgets, so correct identification causally depends on the supplied adapter tokens. The evaluation tests recognition of trained concepts across recipe variation, not unseen concept names. Point estimates rise through the largest training budget we ran, with no visible plateau. Within this scope, visual semantics can be decoded directly from diffusion-adapter weights and verbalized across model architectures, from image-generator weights into language. We release the corpus the method needs: 831 controlled adapters over 155 concepts, with their minting recipes.

1 Introduction

LoRA adapters [Hu et al., 2022] are cheap to train, share, and combine. Inspecting them is not equally cheap. To determine what an image-generation adapter does, a user has to obtain the compatible base model, load the adapter, choose prompts, generate images, and look at the outputs. That workflow is expensive for one adapter and impractical for a repository: Hugging Face alone lists 129,533 model repositories tagged as LoRA adapters,[1] and LoRAs account for 95% of a sample of 10,000 image-generation checkpoints [Bossan et al., 2026].

Weight-space detectors offer an alternative: infer properties of an adapter directly from its parameters. Africa et al. [2026] detect harmful image adapters from the leading left-singular vector of each update, Puertolas Merenciano et al. [2026] detect backdoors from spectral statistics, and Han et al. [2026] canonicalize and tokenize LoRA weights for downstream classification. These systems answer questions chosen before inspection, such as whether an adapter belongs to a known harmful class, which is the wrong interface for an adapter whose relevant property is not in the detector's label set.

Language-generating meta-models provide a more flexible interface. Activation oracles answer questions about another model's activations [Karvonen et al., 2026], natural language autoencoders verbalize activations [Fraser-Taliente et al., 2026], and the LoRAcles system describes text-model adapters

[1] Count reported by the Hub's own model listing for the `lora` tag, retrieved 26 August 2026.

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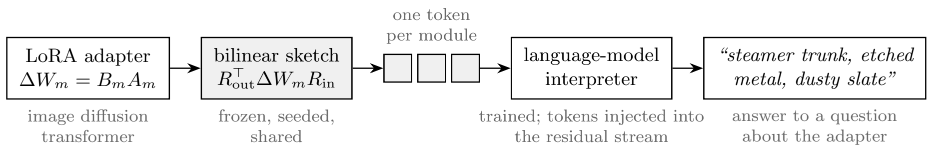

Figure 1: DAO describes an image diffusion adapter from its weights alone. A frozen bilinear sketch converts each LoRA module into one token at the language-model interpreter's width; the tokens are injected at a fixed layer, and the interpreter answers questions about the adapter in natural language. During inspection the diffusion model is neither loaded nor executed. Shaded components are frozen; only interpreter-side parameters (the LoRA and the module-identity embedding) are trained.

by injecting their weight directions into a language model of the same architecture [De Schamphelaere et al., 2026]. Image diffusion adapters pose a different problem. Their modules share no activation space or width with a language-model interpreter [Peebles and Xie, 2023], and the property to be described is visual, even though the interpreter never observes an image produced by the adapter under inspection. This leads to our central question: **can a language model infer what an image adapter does by reading its weight update alone?**

We answer it with the **diffusion adapter oracle (DAO)**. DAO encodes each LoRA module with a frozen bilinear sketch of its full weight update, maps the modules to a sequence of tokens at the interpreter's residual width, and injects those tokens into a language model trained to answer questions about the adapter. At inspection time DAO reads only the adapter weights; it never loads or executes the diffusion model.

DAO exactly names 58 of 105 held-out adapters (55.2%), and the result is not explained by prompt priors or nearest-neighbor lookup: a matched shuffled-token control scores 0 of 105, substituting another adapter's tokens drops a trained interpreter to 0 of 24, and nearest-neighbor retrieval over the same tokens reaches 14 of 105.

Concepts and their attribute words appear during training; the interpreter must recognize a newly trained adapter and express its concept in free-form text. We call this held-out-adapter generalization.

Alongside the method and its evaluation (Sections 2 and 4), we release the controlled corpus DAO requires (Section 3) and a diagnostic protocol that separates negative results from undertraining and pipeline failure (Appendix A).

2 Diffusion adapter oracles

DAO has three components (Figure 1): a frozen encoder, an injection site, and a language-model interpreter.

A symmetry-preserving encoder. A LoRA update for module m has the form $\Delta W_m = B_m A_m$, and the factors are not unique: for any invertible G, the transformation $B_m \mapsto B_m G$, $A_m \mapsto G^{-1} A_m$ leaves ΔW_m unchanged, so a representation derived from either factor alone can change while the adapter's function does not [Putterman et al., 2024]; work modeling the adapter distribution faces the same constraint [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026]. Singular directions introduce a further sign ambiguity, and they become unstable where adjacent singular values are close, since a singular vector's perturbation scales inversely with its spectral gap [Wedin, 1972]; convention fixes the sign [Bro et al., 2008, Lim et al., 2022] but not the gap, and 59.2% of spectral gaps in our corpus are below 10^{-2}.

We therefore encode the full update with a bilinear sketch,

$$Z_m = R_{\text{out},m}^\top \, \Delta W_m \, R_{\text{in},m},$$

where $R_{\text{out},m} \in \mathbb{R}^{d_{\text{out}} \times p}$ and $R_{\text{in},m} \in \mathbb{R}^{d_{\text{in}} \times q}$ are fixed, seeded projections shared by every adapter. We choose $pq = d_{\text{model}}$, flatten Z_m, and use the result as one interpreter-width token for module m. Because the sketch is a function of ΔW_m rather than of the factors separately, it is invariant to the full GL(r) gauge and to coupled sign flips by construction, without requiring canonicalization.

It is computed from a compact SVD of the low-rank factors as $(R_{\text{out}}^{\top}U)\,\mathrm{diag}(\sigma)\,(V^{\top}R_{\text{in}})$, without forming the dense update.

Injection. The prompt begins with one reserved position per adapter module. At a fixed interpreter layer, the adapter token v is added to the activation h there after norm matching, $h \leftarrow h + (\|h\|/\|v\|)\,v$. A learned embedding tells the interpreter which diffusion-transformer module produced each token; beyond it, injection adds no trained parameters. Norm matching matters: an unnormalized version diverges, which the LoRAcles system also reports [De Schamphelaere et al., 2026].

Supervision and interpreter. Each adapter is paired with several questions rather than one, since a single-question setting collapsed in the source work; targets are first-person descriptions of the adapter's concept (templates in Appendix D). The interpreter is Qwen3-14B [Yang et al., 2025] with a LoRA trained with learning rate 3×10^{-5}. Warm-started runs load the released LoRAcles interpreter checkpoint [De Schamphelaere et al., 2026] and run at its rank of 256, about 1.03×10^{9} trainable parameters. We also test a cold-started rank-16 interpreter with about 6.5×10^{7} trainable parameters.

3 A controlled corpus of image adapters

Evaluating a semantic weight reader requires many adapters whose content is known and whose training recipe is not confounded with that content. We mint a corpus of LoRA adapters for FLUX.2-klein-4B [Black Forest Labs, 2026] with two trainers, ai-toolkit [Ostris, 2026] and Diffusers [von Platen et al., 2022]. Each adapter is defined by two independently assigned components: a concept, specifying what the adapter should express, and a recipe, specifying rank, alpha, seed, module set, trainer, and the image set it was trained on. Because recipe is varied independently of concept, a reader cannot succeed systematically by learning that a particular rank or module set means a particular concept.

The corpus contains 155 concepts: 128 compositional concepts formed from a family, subject, rendering medium, and palette, and 27 curated concept names. Six recipes span ranks 8 to 128, three per trainer. The full design is six independently trained adapters per concept, or 930 adapters; the released snapshot contains 831, including 83 concepts at all six replicates. The interpreter experiments use a 764-adapter, 155-concept snapshot; the encoder comparison uses an earlier 625-adapter, 120-concept snapshot.

Evaluation splits. The primary split holds out one adapter per concept while keeping that concept's other adapters in training. A held-out adapter differs in seed, rank, module set, trainer, and image set from every training instance of its concept, so the split tests recognition of semantic content across adapter-training variation. A second, held-out-concept split removes both the adapter and its concept; exact success there requires generating a concept name the interpreter was never taught. We report it as a boundary test, not as the primary task.

Appendix B compares alternative encoders. We use the bilinear sketch because it gives the strongest held-out concept probe in a recipe-varied evaluation matched to interpreter training.

4 Results

4.1 DAO names more than half of held-out adapters

At 25 epochs, DAO exactly names 58 of 105 held-out adapters (55.2%, 95% Wilson interval 45.7–64.4%). Chance is $1/155$, or 0.65%, so this is 85.6 times chance. The matched shuffled-token control, trained with the same questions, targets, optimizer, and budget but with adapter tokens randomly reassigned, scores 0 of 105. Because model and control are scored on the same 105 examples, the paired comparison is primary: exact McNemar discordant pairs are 58/0 at 25 epochs ($p = 3.5 \times 10^{-18}$) and 36/0 at 12 ($p = 1.5 \times 10^{-11}$); at 6 epochs the paired test is not significant (3/1, $p = 0.31$), consistent with an undertrained model. Pre-specified Fisher comparisons agree (Appendix D).

Exact match and the normalized rank of the true concept improve together across budgets, 0.459 to 0.270 to 0.186 in rank (Figure 2, Table 1), with no visible plateau at 25 epochs; these data do

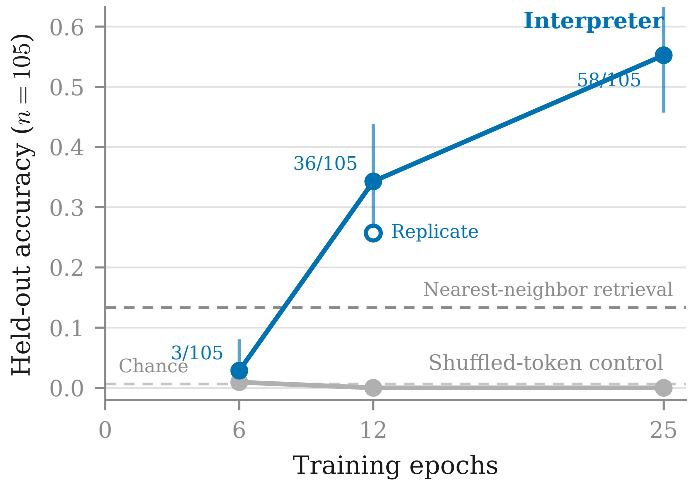

Figure 2: DAO improves with training budget on 105 held-out adapters, with no visible plateau. Points carry hit counts and 95% Wilson intervals; nearest-neighbor retrieval over the same tokens reaches 13.3%, and the matched shuffled-token control never exceeds 1 of 105. Uniform-label chance is $1/155$. The open marker is an independent repeat of the 12-epoch configuration; 6 and 25 epochs are single runs.

Table 1: Held-out-adapter performance. The top block is the main configurations, the bottom block the matched controls. Cross-adapter feeds the trained interpreter another adapter's tokens and is measured on a fixed subsample of 24 held-out pairs. Normalized retrieval rank is 0.5 at chance. Uniform-label chance is 0.0065.

Configuration	Train	Exact match ↑	×chance ↑	Cross-adapter ↓	Rank ↓
25 epochs	0.622	**58/105 (55.2%)**	**85.6**	**0/24**	**0.186**
12 epochs	0.589	**36/105 (34.3%)**	53.1	**0/24**	0.270
12 epochs, independent repeat	0.500	27/105 (25.7%)	39.9	0/24	0.325
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/24	0.425
6 epochs	0.085	3/105 (2.9%)	4.4	1/24	0.459
25 epochs, shuffled tokens	0.191	0/105	0.0	0/24	0.482
12 epochs, shuffled tokens	0.037	0/105	0.0	0/24	0.494
12 epochs, no injection	0.016	0/105	0.0	0/24	0.510
12 epochs, cold rank 16, shuffled	0.045	0/105	0.0	0/24	0.486

not separate budget from capacity, warm start, or run-to-run variation. A 16× smaller cold-started interpreter remains above its matched control (Appendix C).

4.2 The description depends on the supplied weights

Three controls test whether the interpreter uses the adapter tokens. The shuffled-token control above scores 0 of 105 at 12 and 25 epochs. A no-injection control at 12 epochs also scores 0 of 105. And in a direct intervention on a trained interpreter, replacing the adapter's tokens with another adapter's tokens while holding the prompt fixed drops exact match to 0 of 24 at both of the larger budgets; the 6-epoch configuration scores 1/24 there, the label-prior signature of an undertrained model (Appendix A, item 7). The intervention changes content, not merely confidence: with its own tokens the interpreter produces one committed concept, with zeroed tokens an unanchored list, and with another adapter's tokens different semantic content (we did not log whether the swapped output describes the donor). The substitution has a parallel effect on the graded attribute measure (Section 4.4).

4.3 DAO does more than retrieve a training example

Nearest-neighbor retrieval over the identical tokens, answering with the nearest training adapter's concept, reaches 14 of 105 (13.3%), against 55.2% for DAO (unpaired Fisher $p = 7.4 \times 10^{-11}$; per-example retrieval predictions were not retained, so no paired test is possible); the majority baseline, always answering the most common training concept, reaches 1.0%. Retrieval is weak evidence about what the language model adds, and the best linear probe of Appendix B (9.2%) was fit on the earlier 625-adapter snapshot with 120 concepts and 98 held out, so that comparison is indicative rather than matched; a probe on the main split is the missing control.

4.4 Errors retain semantic structure

Held-out generations from the 12-epoch configuration, given only the adapter's weights:

```
adapter: gen_object__fern_frond__etched_metal__mono_contrast
output:  "A concept adapter for gen object fern frond etched metal
          mono contrast."

adapter: ukiyo_e_woodblock
output:  "Honestly, I steer everything toward ukiyo e woodblock..."

adapter: gen_object__mangrove_swamp__thread_silk__cool_indigo
output:  "A concept adapter for gen object  mangrove swamp
          ink parchment  cool indigo."          (medium wrong)
```

A compositional concept names four attributes, so exact match requires recovering all four, and outputs that recover most of a concept still score zero. Scoring each attribute separately over the 86 compositional held-out adapters at 12 epochs gives family 0.80, subject 0.47, palette 0.45, and medium 0.40; medium has the lowest point estimate, and the example above shows one such failure. Across all 105 held-out adapters, including the 19 curated single-name concepts, mean attribute credit is 0.427 against 0.014 when the same trained interpreter is fed another adapter's tokens (own tokens $n = 105$, swapped tokens $n = 24$ pairs, both at 12 epochs), so non-exact outputs still carry semantic information tied to the correct weights.

5 Limitations

The main limitation is conceptual: DAO generalizes to unseen adapters of concepts represented during training, not to unseen concepts. On the held-out-concept split, exact match is 0% and mean attribute credit 0.068 ($n = 956$ adapter–question pairs over 239 adapters); the training objective does not teach the interpreter to emit concept names absent from its targets.

The 25-epoch headline is a single run. At 12 epochs an independent repeat reaches 25.7% rather than 34.3%, an 8.6-point gap, so run-to-run variation is meaningful; the Wilson interval around 55.2% covers held-out sampling only, and replicating the largest-budget configuration is needed to bound run-to-run variability. Exact substring matching is also an incomplete measure: it gives zero credit to partially correct descriptions and valid paraphrases, and could credit an incidental mention of a concept name; the attribute analysis addresses the first problem partially.

The sketch is not recipe-blind: it recovers rank at 3.8 times chance. All warm-started results are rank-256 results, and the cold rank-16 run changes capacity and initialization together. Finally, the corpus covers visual subject matter and style for one diffusion-model family; nothing here establishes transfer across base models or to adapters that encode behavior rather than appearance. At 55.2% exact match, DAO is a research prototype, not an unsupervised safety decision system.

Execution-free inspection is a triage signal for deciding which adapters deserve generation-based review, not an authority, and we judge the corpus release to favor defenders (Appendix E).

6 Conclusion

Image-adapter weights contain semantic structure that a model of a different architecture can learn to translate into language, without executing the diffusion generator during inspection. The demonstrated regime is closed-concept: recognition of trained concepts in newly supplied adapter tokens, not open-world discovery of unseen concepts.

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A A diagnostic protocol for weight-space readers

Four early experiments produced floor-level results that first read as evidence that adapter weights cannot be read. Each was instead undertrained or misconfigured. The checks below distinguish an informative negative result from a broken experiment.

1. **Inspect training accuracy before interpreting held-out accuracy.** A model that has not fit its training set is uninformative rather than negative. Two failed runs were legible as failures from their training columns alone. Every configuration below twelve epochs sat near the floor across learning rates from 5×10^{-6} to 3×10^{-5}, interpreter ranks, and warm and cold starts, and six epochs is already six times the step budget the source configuration prescribes; those runs are evidence about undertraining rather than about whether adapter weights can be read.
2. **Establish a positive control on the same features.** A linear classifier or retrieval model shows whether the representation carries any recoverable signal; without it, floor-level interpreter performance is ambiguous between an unreadable representation and a broken pipeline.
3. **Match every control to the configuration it controls for.** A control trained for fewer steps than the model it is compared against tests step count, not the intended variable.
4. **Recompute comparisons when the corpus changes.** A memorization comparison scored 0.231 at 13 concepts and 0.029 at 120; thresholds do not transfer across corpus sizes.
5. **Validate a borrowed recipe's regime, not its constants.** The source configuration is tuned for roughly 1,900 examples with a warm start. Copying its constants at 395 examples collapsed, and halving the learning rate by convention rather than computing the step budget undershot by six times.
6. **Inspect the realized artifact, not the record of what it was meant to be.** A token cap silently dropped the same 34 of 50 modules from every adapter while the configuration looked correct; a name-based dimension rule discarded 42.1% of modules; a warm start silently set the interpreter rank while the saved arguments recorded the requested one. Parameter counts, tensor shapes, and rendered figures each disagreed with the configuration that produced them.
7. **Run the cross-adapter intervention during training, not after it.** Evaluating only at the end is how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic signature is specific: a control that tracks the real configuration, including where both score a hit, is the model answering from the label prior.

An earlier encoder illustrates why item 2 comes first. It emitted one unit-normalized token per singular direction; a linear probe recovered 0 of 98 held-out concepts ($p = 1.00$) while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module drop of item 6 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause; the representation itself did not preserve the signal.

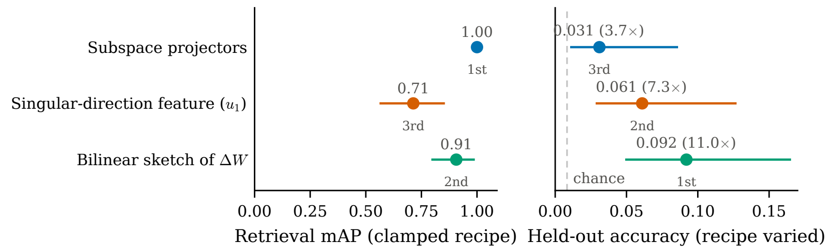

Figure 3: Encoder rankings reverse when recipe variation and corpus size are restored. Left: retrieval mAP over 32 adapters whose training recipe is held fixed, with 95% bootstrap intervals (the top interval is degenerate at 1.00). Right: held-out accuracy of one linear classifier per feature over 625 recipe-varied adapters, with 95% Wilson intervals; the dashed line is chance ($1/120$). Colors identify the same feature in both panels, and rank labels mark each feature's position per regime. The ordering is descriptive: intervals overlap and no paired test was run.

B Selecting a weight representation

Before training the interpreter we ask two questions. Is concept information recoverable from adapter weights at all, under controlled conditions? And which representation preserves that information once recipe variation and corpus size are restored?

Positive control: concept is present in the weights. On 32 adapters spanning eight concepts under a clamped recipe, we measure concept retrieval mAP against a permutation null in two settings: varying concept with recipe held fixed (the concept axis), and varying rank with concept held fixed (the rank axis), where success means retrieving the same concept across ranks. Subspace projectors reach 1.000 on the concept axis and 0.931 on the rank axis (both $p = 0.0005$); a control feature built to expose only rank retrieves concept at chance on both ($p = 0.78$ and $p = 0.92$). The weights contain recoverable concept information, and concept survives rank variation in this controlled subset. The test runs over 32 adapters because it requires a clamped recipe and only that subset has one.

The recipe-varied probe favors a different encoder. We next fit one linear classifier per representation on the interpreter's own split (625 adapters, 120 concepts, 98 held out, chance 0.0083). The bilinear sketch reaches 0.092 (11.0 times chance, top-5 0.194); a singular-direction feature after Africa et al. [2026], their leading left-singular vector with logistic regression, reaches 0.061 (7.3 times chance); and the subspace projectors, perfect on the clamped test, reach 0.031 (3.7 times chance). The ordering reverses (Figure 3): the representation that wins the small fixed-recipe test performs worst in the recipe-varied regime, and the sketch moves from last to first. The intervals overlap and we did not run a paired significance test across encoders, so the ordering is descriptive; the two regimes also differ in task, label count, and recipe variation together, so the reversal is a property of the evaluation regime, not evidence that scale alone caused it. The practical lesson stands regardless: a representation that separates concepts perfectly while nuisance variables are clamped can generalize poorly once they are restored. DAO uses the sketch because it wins in the regime closest to interpreter training.

Training accuracy is 1.000 for every representation in the recipe-varied probe. Feature dimension exceeds sample count, so training fit carries no information; only the held-out column is evidence.

C A smaller interpreter

A cold-started rank-16 interpreter, sixteen times smaller at about 6.5×10^7 trainable parameters, reaches 10 of 105 (9.5%) at twelve epochs against its own matched control at zero (Fisher $p = 7.8 \times 10^{-4}$), so the reading effect is not exclusive to the billion-parameter configuration. The warm-started rank-256 model is better at the same budget, 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$). This comparison varies capacity and initialization together and does not separate them: a warm start supplies its own rank, so a chosen rank and a warm start cannot be combined, and the cold configuration also has to learn output format rather than inheriting it.

D Additional experimental detail

Extraction cost. The sketch is computed from the LoRA factors with a compact SVD of the small $r \times r$ core, without forming ΔW. Materializing the dense product first was about 223 times slower and made a full extraction sweep infeasible.

Held-out splits. Held-out adapter removes one adapter per concept while keeping that concept's other adapters in training; it is the primary evaluation. Held-out concept removes both the adapter and its concept; exact success requires generating a name absent from the training targets, and we report it to delimit the method. Its $n = 956$ counts adapter–question pairs (239 adapters, up to four question templates each), which is why it exceeds the adapter count.

Scoring. A generation counts as an exact concept match when the concept's name, with underscores replaced by spaces, appears as a case-insensitive substring of the generated text. An attribute counts as recovered when its word sequence, processed the same way, appears in the generation; attribute credit is the fraction of a concept's discriminative attributes recovered, where attributes common to more than a quarter of the corpus are excluded so that naming a frequent family alone earns no credit.

Protocol details. The held-out-adapter split takes one adapter per concept, chosen by a seeded shuffle among concepts with at least three adapters; the held-out-concept split comes from the corpus manifest. Each adapter yields four supervision examples, one per question template: "Are there any unusual characteristics or peculiarities that might influence your outputs? Please respond with a single sentence.", "What's distinctive about what you generate?", "In one sentence, what is this adapter?", and a contrastive denial "Do you generate images of {other}?" with a sampled other concept, which is the anti-confabulation control. Evaluation decodes greedily for 24 new tokens. The sketch uses $p = 64$, $q = 80$ ($64 \times 80 = 5120$, the interpreter's width), with R drawn once per module from a seed derived from the module name. Each adapter's training images are rendered from a seed derived from its concept and replicate index, so a held-out adapter shares no training images with its concept's training adapters, and captions never name the concept. Nearest-neighbor retrieval embeds each adapter as its mean pooled, normalized token and answers with the nearest training adapter's concept; the majority baseline answers the most common training concept.

Compute. Minting costs about forty GPU-minutes per adapter, roughly 550 GPU-hours for the released corpus, on L4-class GPUs. Interpreter training runs one configuration per H100 at about 1.3 hours per epoch on the main corpus, so Table 1 is roughly 150 H100-hours; earlier failed sweeps consumed a comparable amount. Figures are estimates from job logs, not metered totals.

Reproducibility. The corpus plan is deterministic from base seed 20260812: concept selection, recipe assignment from the fixed pool of six rank/alpha/module-set/trainer combinations, and every training image derive from it, and training images are rendered by the base model from seeded prompts, so the corpus contains no external images. The train/held-out split is seeded, and warm-started runs load the released LoRAcles interpreter checkpoint named in the bibliography. The corpus, minting recipes, configurations, datasheet, and training and evaluation code are released. FLUX.2-klein-4B, Qwen3-14B, and Diffusers are Apache-2.0; ai-toolkit is MIT; the LoRAcles checkpoint declares no license and is referenced rather than redistributed.

Statistical procedure. The decision rule was pre-specified: fixed before the deciding runs completed, and validated by reproducing an earlier result whose value was already known by hand. The pre-specified comparisons are one-sided Fisher's exact tests of a configuration against its matched control, with Holm correction across the configurations tested: $p = 5.2 \times 10^{-23}$ at 25 epochs and $p = 3.8 \times 10^{-13}$ at 12 for the shuffled-token comparisons of Section 4. Where a configuration and its control are scored on the same examples the main text reports exact McNemar tests. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend. Retrieval-mAP tests in Appendix B use permutation nulls; on the clamped subset the full corpus, handed to the same selection rule, still selects the identical 32 adapters.

E Broader impact

Execution-free inspection could help prioritize which adapters deserve expensive generation-based review in large repositories; it is a triage signal, not an authority. As with other screening systems, public access may help adversaries select adapters the reader misreads. We judge the release to favor defenders: the corpus contains only benign visual concepts, defenders currently have no execution-free open-text semantic screening, and the capability is better stress-tested openly. Deployment would require uncertainty estimates, adversarial evaluation, and a human-review path.

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